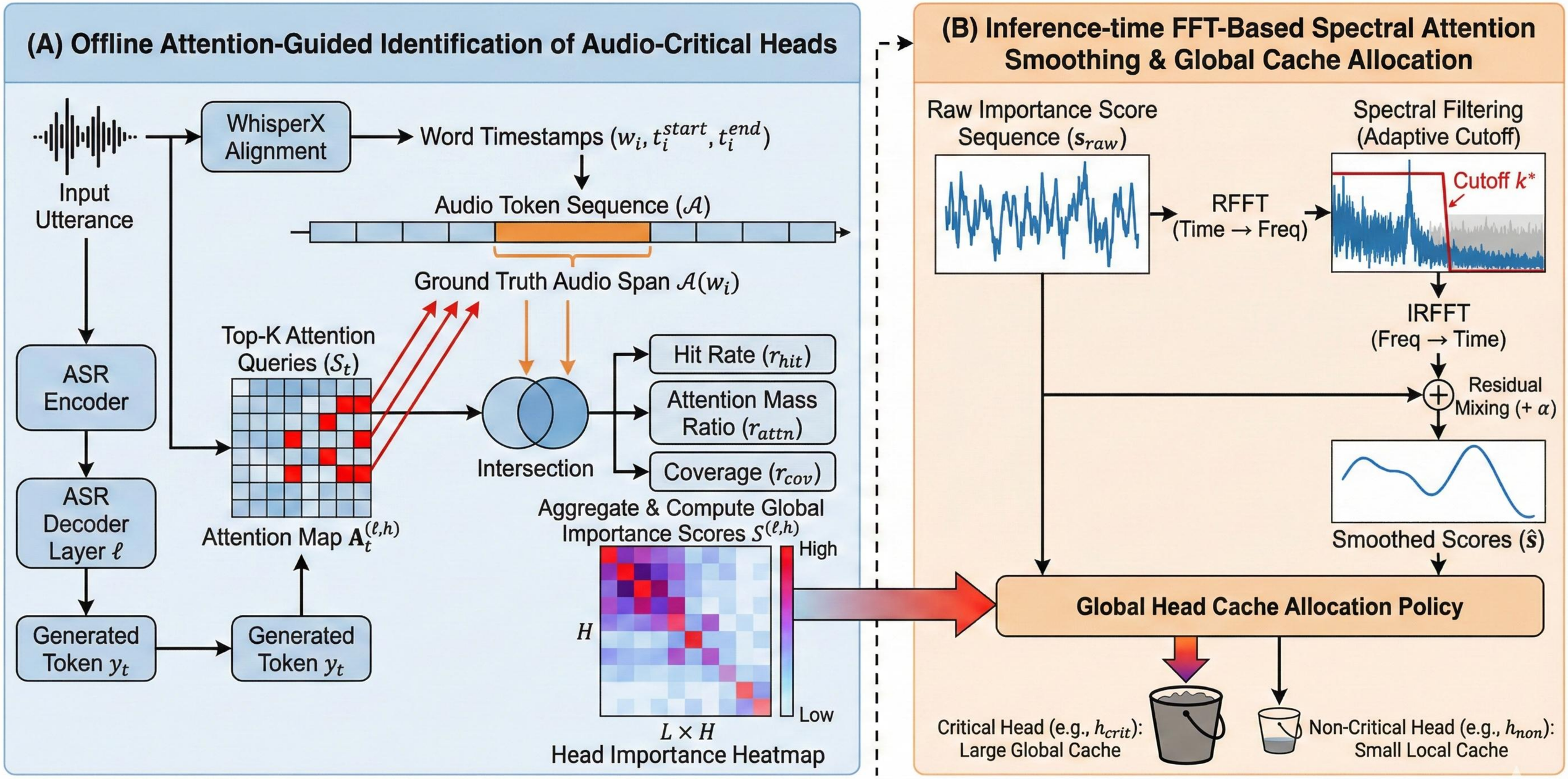

(A) Offline Attention-Guided Identification of Audio-Critical Heads
Input Utterance
WhisperX Alignment
Word Timestamps $(w_i, t_i^{start}, t_i^{end})$
Audio Token Sequence $(\mathcal{A})$
Ground Truth Audio Span $\mathcal{A}(w_i)$
ASR Encoder
ASR Decoder Layer ℓ
Generated Token y_t
Generated Token y_t
Top-K Attention Queries (S_t)
Attention Map $\mathbf{A}_t^{(\ell,h)}$
Intersection
Hit Rate (r_{hit})
Attention Mass Ratio (r_{attn})
Coverage (r_{cov})
Aggregate & Compute Global Importance Scores $S^{(\ell,h)}$
High
Low
H
$L \times H$
Head Importance Heatmap
(B) Inference-time FFT-Based Spectral Attention Smoothing & Global Cache Allocation
Raw Importance Score Sequence (s_{raw})
RFFT (Time → Freq)
Spectral Filtering (Adaptive Cutoff)
Cutoff k^*
IRFFT (Freq → Time)
Residual Mixing (+ α)
Smoothed Scores $(\hat{s})$
Global Head Cache Allocation Policy
Critical Head (e.g., h_{crit}): Large Global Cache
Non-Critical Head (e.g., h_{non}): Small Local Cache